

torch.take_along_dim#

https://pytorch.org/docs/stable/generated/torch.take_along_dim.html

Description:

Selects values from input at the 1-dimensional indices from indices along the given dim. If dim is None, the input array is treated as if it has been flattened to 1d. Functions that return indices along a dimension, like `torch.argmax()` and `torch.argsort()`, are designed to work with this function. See the examples below. This function is similar to NumPy's `take_along_axis`. See also `torch.gather()`.

Function Signature

```
torch.take_along_dim(input, indices, dim=None, *, out=None)  
→ Tensor#
```

Parameters

Keyword Arguments

Parameters

Keyword

out (Tensor, optional) – the output tensor.

Code Examples

```
>>> t = torch.tensor([[10, 30, 20], [60, 40, 50]])
>>> max_idx = torch.argmax(t)
>>> torch.take_along_dim(t, max_idx)
tensor([60])
>>> sorted_idx = torch.argsort(t, dim=1)
>>> torch.take_along_dim(t, sorted_idx, dim=1)
tensor([[10, 20, 30],
        [40, 50, 60]])
```

Notes & Warnings

NOTE This function is similar to NumPy's `take_along_axis`. See also `torch.gather()`.

Detailed Documentation

Documentation

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If dim is None, the input array is treated as if it has been flattened to 1d.

Functions that return indices along a dimension, like `torch.argmax()` and `torch.argsort()`, are designed to work with this function. See the examples below.

This function is similar to NumPy's `take_along_axis`. See also `torch.gather()`.

input (Tensor) – the input tensor.

indices (LongTensor) – the indices into input. Must have long dtype.

dim (int, optional) – dimension to select along. Default: 0

out (Tensor, optional) – the output tensor.